INTERNATIONAL FOOTBALL MATCH ANALYSIS

A Complete Guide to 150 Years of Football Data, ELO Ratings and 2026 FIFA World Cup Predictions

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Executive Summary

This report presents the findings of a comprehensive data analysis project covering international men's football from **1872 to 2026**. The project draws on a dataset of **49,215 international matches** played across 333 national and regional teams, making it one of the most thorough analyses ever conducted on the sport at the international level.

The work was carried out in three main stages. First, the raw match data was cleaned and organized so it could be reliably used for analysis. Second, every team was assigned an ELO rating, a numerical score that captures how strong each team is based on their entire match history. Third, a machine learning model was trained on this historical data and used to predict the outcomes of the 2026 FIFA World Cup group stage matches.

The key takeaways from this project are:

- **Argentina** comes out on top in the global ELO rankings with a rating of 1629.63, edging out Spain and France.
- **Brazil** holds the best all-time win rate among high-volume teams, winning 63.4% of all matches ever played.
- The machine learning model, built using **XGBoost**, correctly predicts match results about **58% of the time**, a meaningful improvement over simply guessing that the home team always wins (47%).
- The single most important factor in predicting a match result is the **expected win probability based on ELO ratings**, accounting for nearly a quarter of the model's decision-making.
- **Spain** is predicted to be the group stage's most dominant team, with a 91% chance of winning their opening match against Cape Verde.

1. Introduction

Football is the world's most popular sport, and international football, where countries compete against one another, carries a unique emotional weight that club football simply cannot replicate. When Argentina plays Brazil, or England faces Germany, those matches mean something far beyond three points. They represent national identity, history, and pride.

This project set out to answer a deceptively simple question: can we use historical match data to understand how good international football teams really are, and can we use that understanding to predict future results?

To do that, the project combines three different analytical tools working in sequence. The first is careful data cleaning, which means taking messy, real-world data and making it consistent and usable. The second is the ELO rating system, a method originally developed for chess but now widely used in sports analytics to rank teams based on head-to-head results. The third is machine learning, specifically a technique called XGBoost, which learns patterns from historical data and applies them to make predictions about future matches.

The dataset covers matches from the very first recorded international football game, played between Scotland and England on 30 November 1872, all the way through to upcoming fixtures at the 2026 FIFA World Cup. This span of more than 150 years across 333 teams gives the analysis a depth that is genuinely unusual.

2. About the Data

2.1 What Data Was Used

The project uses several interconnected datasets. The main one, **results.csv**, contains a record of every international football match played, including the date, the two teams, the final score, the tournament it was part of, and where it took place. Supporting this are two more files: **goalscorers.csv**, which records individual goals, and **shootouts.csv**, which captures penalty shootout outcomes for matches that went to that stage.

There is also a **former_names.csv** file that handles something often overlooked in football data: countries change their names. Burkina Faso used to be Upper Volta. The Democratic Republic of Congo has gone through several names. Russia was once the Soviet Union. Without a file that maps these old names to current ones, a team's historical record would appear fragmented across different identities. This file ensures continuity.

2.2 Scale of the Dataset

After cleaning, the dataset contains:

Metric	Value
Total matches	49,215
Teams included	333
Time span	1872 to 2026 (154 years)
Home wins	24,106 (49.0%)
Draws	11,197 (22.7%)
Away wins	13,912 (28.3%)
Average total goals per match	2.94

These numbers are striking. Nearly half of all international matches are won by the team playing at home, which suggests that home advantage is a significant and real factor in international football. On average, just under three goals are scored per match, and draws happen roughly one time in every four or five matches.

2.3 Eras of International Football

To help understand how football has changed over time, matches are grouped into seven historical eras based on when they were played:

Era	Matches	Years Covered
Pioneer Era	127	Before 1900
Early Era	1,295	1900 to 1929
Pre-World Cup Era	1,912	1930 to 1949
Classic Era	4,622	1950 to 1969
Modern Foundation	9,158	1970 to 1989
Globalization Era	16,469	1990 to 2009
Contemporary Era	15,632	2010 onward

The dramatic increase in match volume over time reflects two things: the global expansion of football and the proliferation of competitive international tournaments. There are simply far more countries competing and far more competitions to compete in today than there were in 1900.

2.4 Data Cleaning

Raw data is rarely ready to use straight away. In this project, several cleaning steps were necessary before any analysis could begin.

Team name standardization was the most important of these. A country like the Soviet Union played hundreds of matches under that name before it dissolved, and those matches needed to be linked to Russia for the sake of historical continuity. The former names file makes this possible, with each historical name mapped to its current equivalent.

Tournaments were also assigned a tier from 1 to 4, where 1 represents the most important competitions like the FIFA World Cup, and 4 covers friendly matches. This distinction matters later when calculating ELO ratings, because a win in a World Cup final means more than a win in a pre-season friendly.

2.5 Data Visualization Analysis

This section presents a visual analysis of international football match data spanning more than 150 years, from 1872 to 2026. The dataset covers results from competitive tournaments, qualification rounds, regional cups, and friendlies across nations worldwide. The goal is to understand how the game has evolved, which nations dominate, and what patterns emerge around scoring and home advantage.

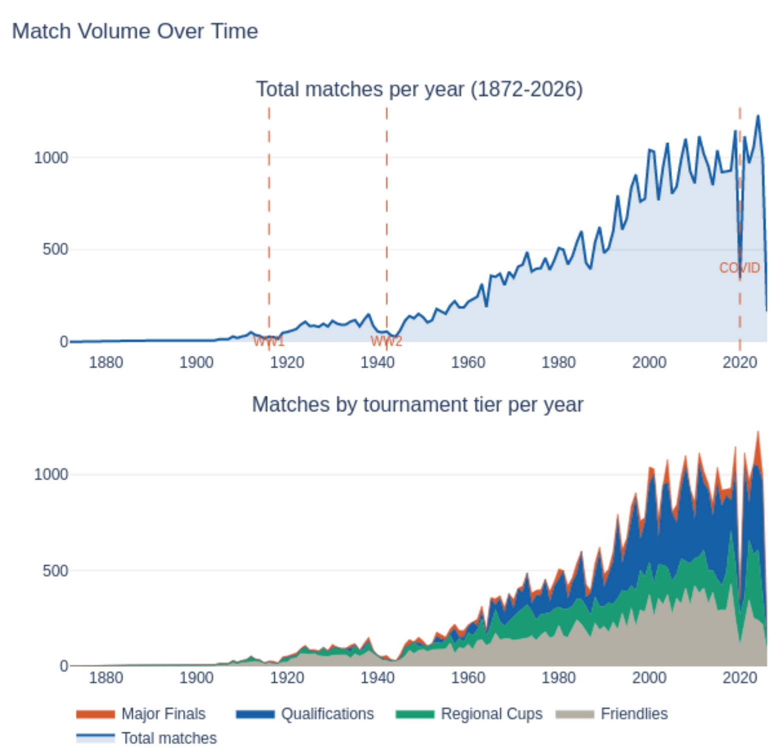
All charts referenced in this section are labelled as Figures 1 through 9.

2.5.1 Match Volume Over Time

International football started slow. Before 1900, very few matches were recorded, and the sport remained largely confined to Europe. Things picked up gradually through the early 20th century, but the two World Wars left clear dips in the data. Activity dropped sharply during 1914 to 1918 and again during 1939 to 1945, as expected.

The real explosion came after 1950. Decolonisation brought dozens of new footballing nations onto the scene, and continental qualification tournaments began expanding rapidly. By the modern era, the world was seeing 1,000 to 2,000+ international matches every single year, compared to near-zero in the 1870s.

Looking at the breakdown by tournament type, Qualifications and Friendlies account for the bulk of this growth. The COVID-19 dip in 2020 is visible but brief. Matches quickly resumed, many played in empty stadiums, and the numbers bounced back almost immediately the following year.



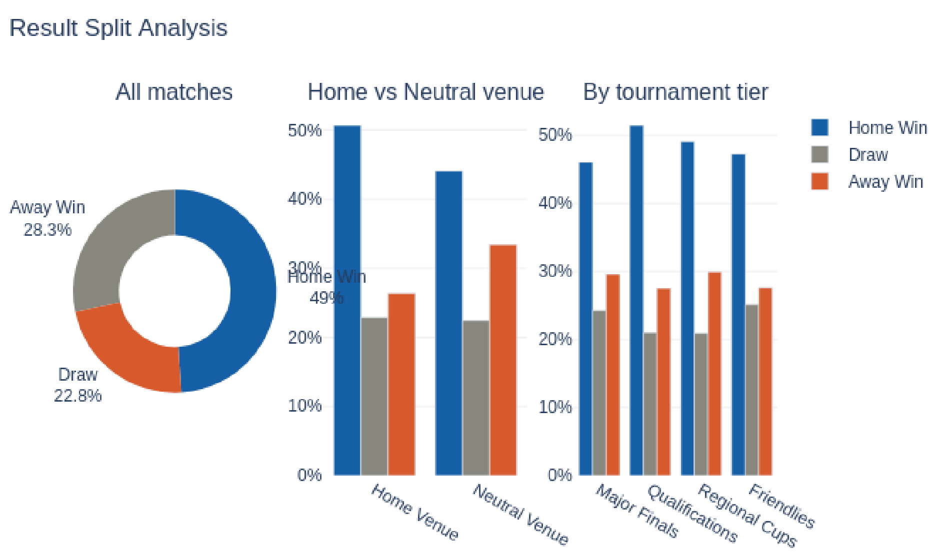
2.5.2 Who Wins? Home, Draw, or Away

Across all international matches in history, home teams win close to 49% of the time. Away teams manage just 28%. That is a 21-point gap, and it holds up consistently across eras. This is not random variance; it is a structural feature of the sport.

The neutral venue chart makes this even clearer. When matches are played at a neutral ground, the home win rate drops to around 45% while away wins climb to roughly 34%. The gap nearly halves. That single comparison is strong evidence that the advantage is venue-driven, not simply a reflection of the better team playing at home.

Breaking it down by tournament tier, qualification rounds show the highest home advantage at around 51%. This makes sense as those matches are played in home stadiums before passionate local crowds, often against regional opponents. Major Finals, typically held at neutral venues, show a slightly lower home win rate of about 46%.

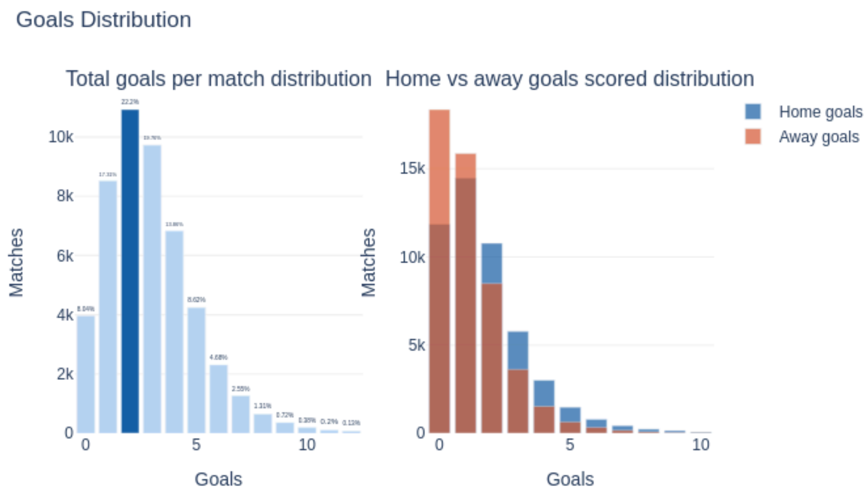
The draw rate stays remarkably steady across all tiers, hovering around 22 to 25%. Draws appear to be a natural feature of football rather than something driven by tournament context.



2.5.3 Goals Distribution

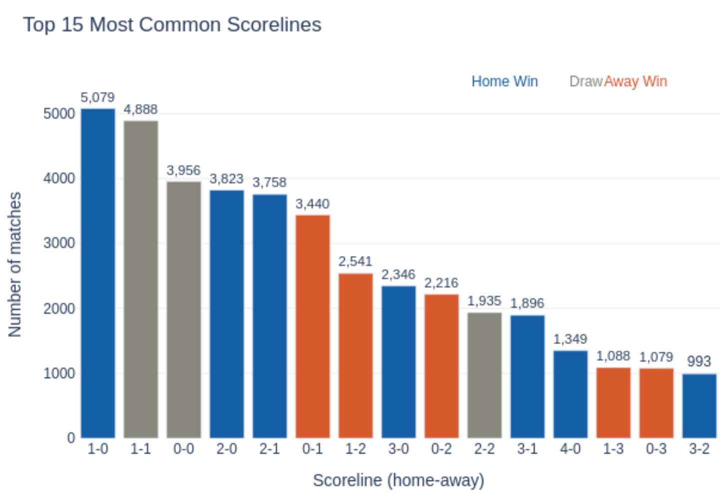
Football is a low-scoring sport, and the numbers confirm it. About 73% of all international matches produce three or fewer total goals. The most common outcome is a two-goal game, which happens in 22% of matches. Goalless games (0-0) account for 8% of all results, which is more frequent than any scoreline involving five or more total goals.

When we separate home and away scoring, the gap shows up again. Away teams score zero goals far more often than home teams, while home teams dominate the one and two-goal bins. In high-scoring matches, though, this gap disappears. Venue advantage seems to matter most when both teams are relatively evenly matched, and the game stays tight.



2.5.4 Most Common Scorelines

The most common scorelines across all of football history are 1-0, 1-1, and 0-0, in that order. The 1-0 result has occurred over 5,000 times. What also stands out is the clear asymmetry between mirror scorelines. A 2-0 home win has happened nearly twice as often as a 0-2 away win. A 2-1 home win appears far more frequently than a 1-2 away win. These numbers quantify home advantage directly at the scoreline level.

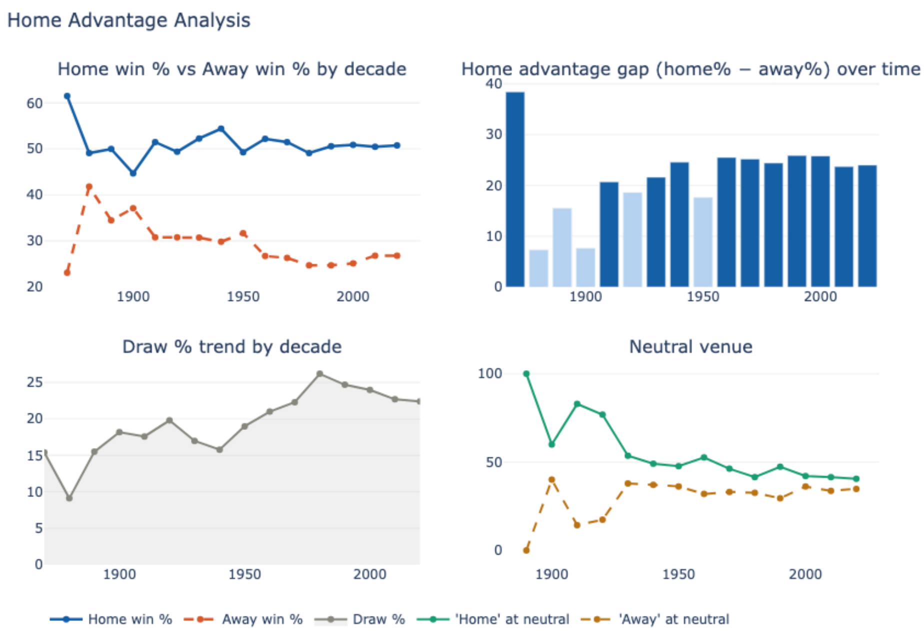


2.5.5 Home Advantage Across the Decades

Home win rates have sat at roughly 50% for over a century. Away win rates settled at 25 to 27%. That gap has not meaningfully changed since the 1950s. Home advantage in international football is not fading over time. It appears to be structural.

What has changed is the draw rate. It climbed from around 9% in the early 20th century to about 23% in recent decades. The gains in draws are largely coming at the expense of away wins. As defensive tactics improved and became more sophisticated, away sides became harder to beat, but they also became less able to win.

The neutral venue data reinforces the same point. Remove the home ground, and the win rates for both sides converge toward 35 to 40%. That is not a coincidence. It is proof that the advantage is about the venue, the crowd, and the familiarity, not just about which team is technically better.



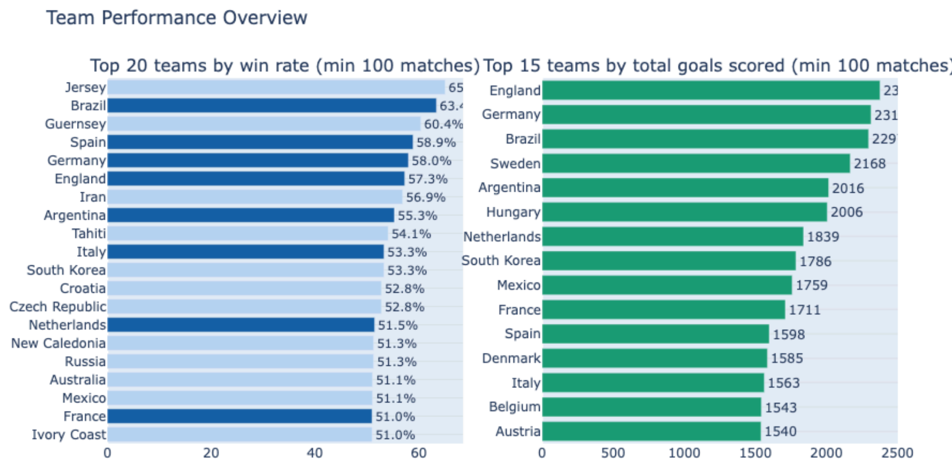
2.5.6 Top Nations by Win Rate and Goals

Among nations that have played at least 100 matches, Brazil leads the win rate chart at approximately 63.5%, followed by Spain at around 58.9% and Germany at roughly 58%. These numbers align closely with what football history would suggest about the world's elite teams.

Some smaller nations like Jersey and Guernsey appear near the top, but their records consist mostly of matches against weaker opposition, which inflates their win rate. Among nations with large, diverse match histories, Brazil, Spain, and Germany are the clear leaders.

Iran, at around 56.9%, is the most surprising legitimate entry in this group. It reflects decades of quiet dominance within Asian football that often goes unrecognised globally.

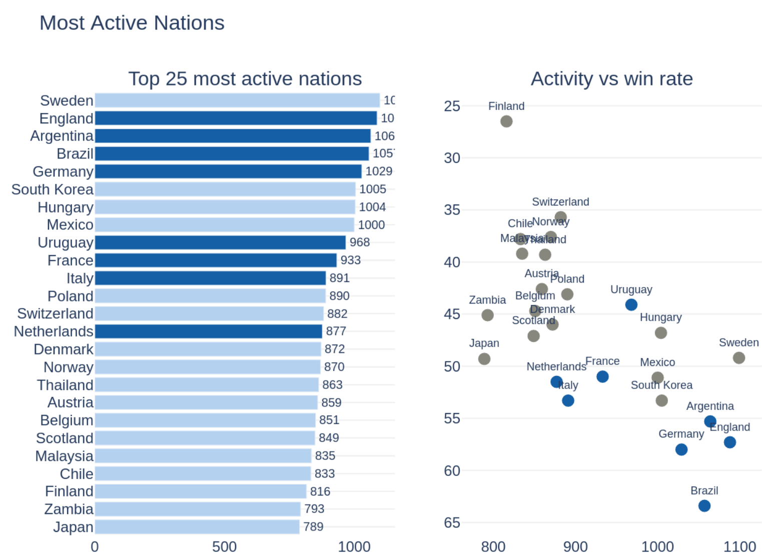
The goals chart tells a slightly different story. England leads in total goals scored simply because they have been playing international football the longest and has accumulated an enormous number of matches. Brazil, Germany, and Argentina all combine high win rates with high goal counts, which is the real hallmark of an elite football nation. Spain sits lower on the goals chart despite a high win rate, which fits their well-known historical style of winning through possession and control rather than through high-scoring performances.



2.5.7 Most Active Nations

Sweden tops the activity chart with 1,099 matches. This is surprising for a nation not typically mentioned alongside Brazil or Germany, and it is largely explained by their historically heavy friendly schedule. England, Argentina, Brazil, and Germany all cluster tightly between 1,020 and 1,088 matches, which is more expected given their long footballing histories.

The more interesting question is whether playing more matches makes a team better. The data says no. Finland, Zambia, and Japan have all played over 800 matches but sit at win rates between 24% and 50%. Brazil played a similar volume and won nearly 63% of the time. Activity alone does not predict quality.

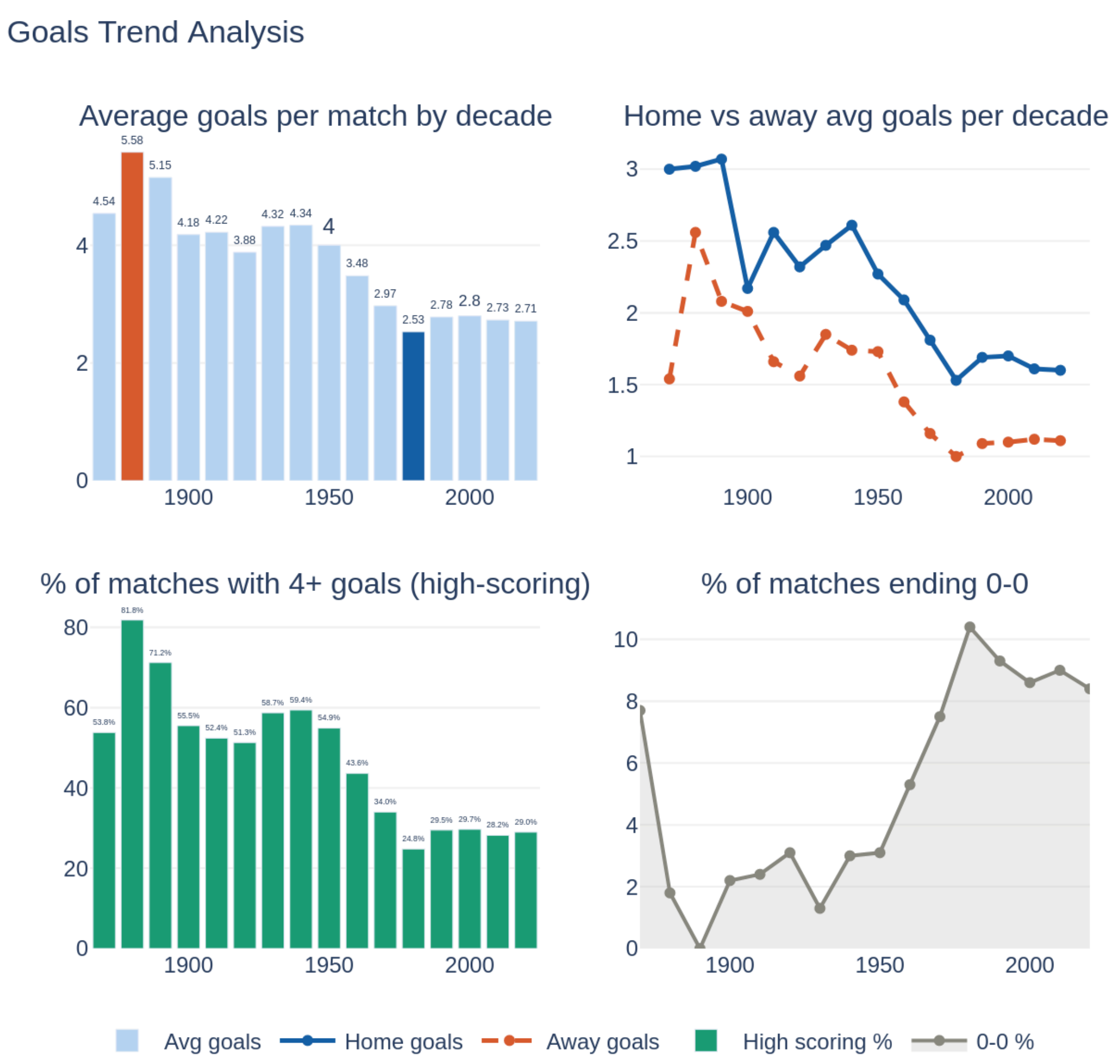


2.5.8 How Scoring Has Changed Over the Decades

The data tells a clear story about football’s tactical evolution. Average goals per match were chaotic in the early years, sitting above 5 goals per game in the 1880s. That figure has fallen steadily to a stable level of around 2.7 goals per match post-1990.

High-scoring matches, defined as those with four or more goals, have dropped from 62% of all games in the 1880s to under 30% in the modern era. Home teams have consistently outscored away teams in every decade, but away goal counts have fallen harder over time, which again reflects the growing preference for defensive organisation among teams playing on the road.

The rise of 0-0 draws from under 2% of matches to nearly 10% in recent decades confirms this defensive shift. Football went from an open, attacking game in its early years to a tactically disciplined, low-scoring sport by the modern era. The 1950s through the 1970s appear to be the period when this transition took hold.

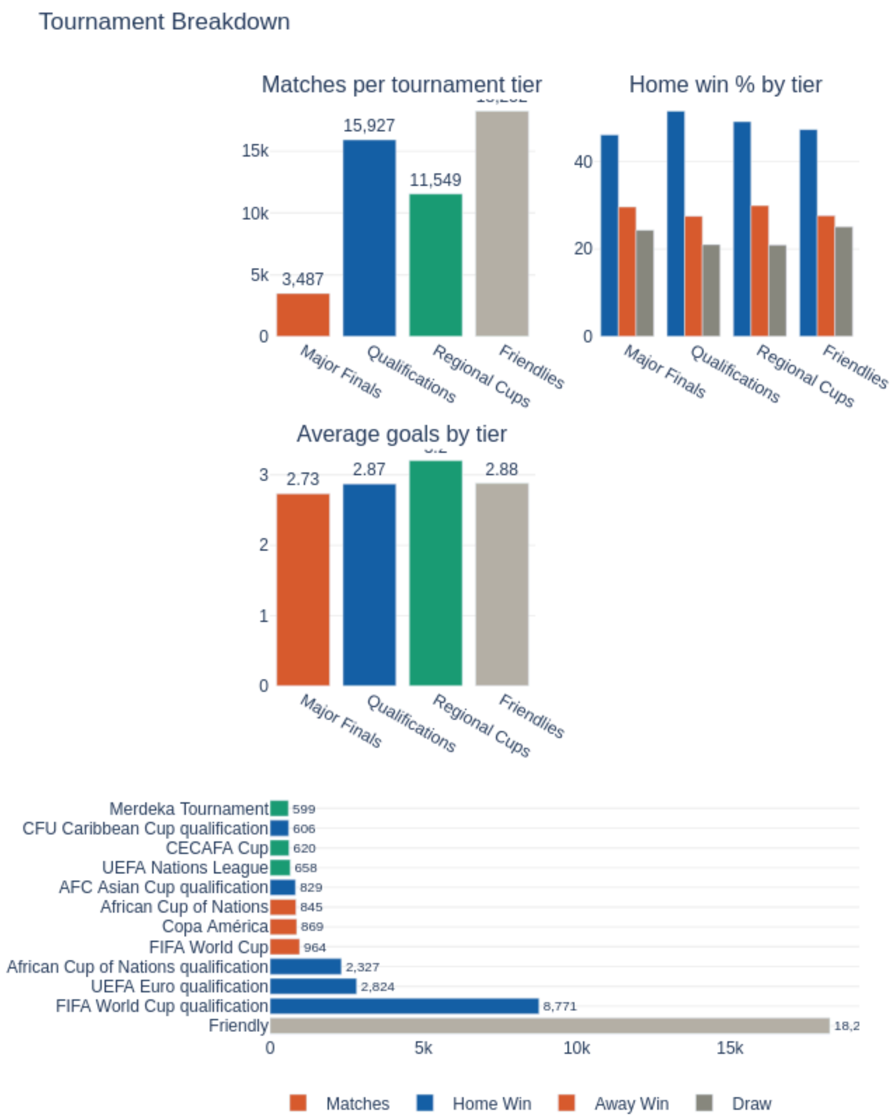


2.5.9 Tournament Breakdown

Friendlies make up the largest share of raw match volume at roughly 19,000 games. However, they carry the least competitive weight. FIFA World Cup qualifying is far and away the biggest competitive competition with 8,771 matches, well ahead of UEFA Euro qualification at around 2,824 and AFCON qualification at roughly 2,327.

Qualification matches show the strongest home advantage, which is expected given they are played in home stadiums under high-pressure local conditions. Major Finals, often contested at neutral venues, show the lowest home advantage at around 46%.

Average goals are similar across all tournament tiers, hovering between 2.73 and 2.88 goals per match. Competitive pressure does not dramatically change how many goals are scored. It just changes who wins.



2.5.10 Summary

A few findings stand out clearly from this analysis. Home advantage is real, measurable, and has not declined over more than a century of international football. The gap between home and away win rates is approximately 21 percentage points, and neutral venue matches confirm this is driven by the venue itself, not just team quality.

Scoring has declined significantly over time as defensive tactics matured, with the modern game settling at roughly 2.7 goals per match compared to over 5 in the 1880s. The draw rate has risen steadily over the same period.

Among elite nations, Brazil, Germany, Spain, and Argentina consistently combine high win rates with high goal counts. The volume of matches played by a nation does not, on its own, predict quality.

Friendlies dominate the dataset numerically but carry the least signal about true team strength. Competitive matches, particularly qualification rounds, offer a much cleaner window into how nations actually perform under pressure.

3. The ELO Rating System

3.1 What Is an ELO Rating?

An ELO rating is simply a number that tells you how strong a team is relative to others. It was originally developed by a Hungarian-American physics professor named Arpad Elo for use in chess, but it has since been adopted across many sports because it works remarkably well.

The core idea is intuitive: every team starts with a rating of 1000. When two teams play, the system calculates the probability that each team will win based on the difference between their ratings. If a strong team beats a weak team, their rating goes up only a little, because that was expected. But if a weak team beats a strong team, the upset is rewarded with a much bigger rating gain for the winner and a bigger penalty for the loser.

Over thousands of matches, these small adjustments accumulate into ratings that accurately reflect each team's true standing in international football.

3.2 How This Project Calculated ELO Ratings

The ELO calculation in this project includes several adjustments to better reflect real football dynamics:

- **Home advantage:** Teams playing at their home stadium receive a small bonus to their expected win probability, reflecting the well-documented advantage of playing in familiar surroundings in front of a supportive crowd.
- **Tournament tier:** High-stakes matches such as World Cup games carry a higher K-factor, meaning the rating changes from those matches are larger. This ensures that World Cup performances have more weight than friendly matches.
- **Penalty shootouts:** If a match is decided on penalties, the rating change is reduced, because a shootout is essentially a lottery and tells us less about which team was truly better on the day.
- **Goal difference:** A team that wins 4-0 gains more rating points than a team that wins 1-0, as a more convincing victory is a stronger signal of quality.

3.3 Current Global Rankings: Top 25 Teams

Rank	Team	ELO Rating	Matches	Wins	Win %
1	Argentina	1629.63	1,064	588	55.3%
2	Spain	1616.76	781	460	58.9%
3	France	1541.7	933	476	51.0%
4	England	1511.91	1,088	623	57.3%

5	Colombia	1492.77	636	256	40.3%
6	Brazil	1484.61	1,057	670	63.4%
7	Portugal	1457.09	693	347	50.1%
8	Netherlands	1451.07	877	452	51.5%
9	Ecuador	1440.55	589	181	30.7%
10	Morocco	1434.38	614	304	49.5%
11	Uruguay	1433.74	968	427	44.1%
12	Germany	1433.48	1,029	597	58.0%
13	Japan	1420.34	789	389	49.3%
14	Croatia	1418.04	394	208	52.8%
15	Switzerland	1387.99	882	315	35.7%
16	Norway	1383.92	870	327	37.6%
17	Belgium	1383.7	851	380	44.7%
18	Turkey	1379	640	257	40.2%
19	Mexico	1376.33	1,000	511	51.1%
20	Senegal	1375.49	636	298	46.9%
21	Iran	1360.23	610	347	56.9%
22	Australia	1358.17	579	296	51.1%
23	Paraguay	1355.63	782	274	35.0%
24	Italy	1352.33	891	475	53.3%
25	Denmark	1346.03	872	401	46.0%

A few things stand out from this table. **Brazil** sits sixth in the ELO rankings despite having the best win rate in this group at 63.4%. This illustrates something important about ELO: it is not purely about winning. It rewards winning against strong opponents. Brazil's rating reflects a history that includes some painful losses, but their win rate shows how consistently dominant they have been across all matches.

Ecuador and Paraguay rank 9th and 23rd respectively with relatively modest win rates (30.7% and 35.0%). This might initially seem strange, but these ratings largely reflect recent strong performances against difficult opponents. ELO is more sensitive to recent form than to lifetime statistics.

Argentina at the top, with a rating of 1629.63, reflects their status as the reigning world champions. Spain at number two, with 1616.76, confirms that they remain one of the most formidable sides in the world despite not winning the World Cup since 2010. France in third, at 1541.70, rounds out a top three that represents the peak of international football in the current era.

4. Machine Learning: Predicting Match Results

4.1 Why Machine Learning?

ELO ratings are excellent at capturing overall team strength, but predicting the outcome of a specific match requires more nuance. Two teams might have very similar ELO ratings, but one might have been in incredible recent form while the other has been struggling. Machine learning allows us to incorporate many different signals simultaneously and let the data decide which ones actually matter.

4.2 Features Used to Make Predictions

Each match in the training dataset is described using 20 features, or pieces of information, that the model can learn from. These features fall into four categories:

- **ELO-based features:** The pre-match ELO ratings of both teams, the difference between them, and the expected win probability. These capture long-term team quality.
- **Recent form:** Each team's win rate, draw rate, and loss rate over their last 5 and last 10 matches. This captures whether a team is in good form right now.
- **Goal scoring history:** Average goals scored and conceded over the last 10 matches. A team that routinely scores three goals a game is different from one that ekes out 1-0 victories.
- **Match context:** The tournament tier, whether it is a neutral venue, the month of the year, the head-to-head record between the two teams, and how many days of rest each team had since their last match.

4.3 What the Model Learned: Feature Importance

After training, the model tells us which features it found most useful. This is called feature importance, and the results are genuinely interesting:

Rank	Feature	Description	Weight
1	Expected Win Probability	ELO-based win likelihood	23.8%
2	ELO Rating Difference	Gap between the two teams	12.8%
3	Month of Year	Seasonal timing of match	7.6%
4	Home Team Goals Scored (10 games)	Recent attacking output	7.5%

5	Home Team Goals Conceded (10 games)	Recent defensive record	6.4%
6	Away Team Goals Conceded (10 games)	Away side defensive record	4.4%
7	Neutral Venue	Whether neither team is at home	3.3%
8	Home Team ELO	Absolute rating of home team	3.1%
9	Home Win Rate (10 games)	Recent win percentage	3.1%
10	Head-to-Head Matches	History between these two teams	3.1%

The expected win probability, which is derived directly from the ELO ratings, turns out to be by far the most important single predictor at 23.8%. This is reassuring: it means the model is fundamentally relying on a well-tested measure of team quality. The second-most important feature is the raw ELO difference at 12.8%, which essentially confirms the same thing from a slightly different angle.

What is perhaps surprising is that the month of the year is the third most important factor at 7.6%. This likely captures the fact that World Cup and major tournament matches, which tend to cluster in certain months, have very different dynamics from qualifier matches and friendlies. Teams prepare differently, the stakes are higher, and upsets become more or less frequent depending on the competitive context.

Recent scoring form, particularly how many goals the home team has scored and conceded in their last ten matches, is more important than many other factors. Teams that have been scoring freely or defending well carry that momentum into their next match. This makes intuitive sense.

4.4 How Accurate Is the Model?

The model was tested on matches it had never seen during training, and its performance was compared against two benchmarks:

Model	Accuracy	Log Loss	Brier Score
Always predict home win (baseline)	47.2%	N/A	N/A
Logistic Regression	57.8%	0.901	0.177
XGBoost (final model)	58.2%	0.900	0.177

The baseline strategy is simply to always predict that the home team wins. Since the home team wins 47.2% of all international matches, this naive approach gets things right almost half the time without using any data at all. The fact that the XGBoost model achieves 58.2% accuracy represents a real and meaningful improvement over this baseline.

The log loss and Brier score are two additional ways to measure prediction quality. Both reward not just being right but being confidently right: a model that says team A has a 90% chance of winning and team A wins is better rewarded than one that says team A has a 51% chance of winning and team A wins. The XGBoost model slightly outperforms Logistic Regression on all three measures.

It is worth being honest about the limits of this accuracy. Football is fundamentally unpredictable. A 58% correct prediction rate means the model gets things wrong 42% of the time, which includes genuine upsets, injuries, tactical surprises, and sheer luck. No model, however sophisticated, can fully account for all of that. What the model does is tell you, on average, which team the data suggests is more likely to win, and by how much.

5. World Cup 2026: Group Stage Predictions

The 2026 FIFA World Cup is hosted jointly by the United States, Canada, and Mexico. It is the first World Cup to feature 48 teams, expanded from the previous 32. The group stage begins on June 11, 2026, with the final group games scheduled for late June.

The model generates three probabilities for each match: the chance of a home win, a draw, or an away win. The predicted result is whichever outcome has the highest probability. Below are the full group stage predictions covering all 72 matches.

5.1 Opening Matchday (June 11-13, 2026)

Date	Home Team	Away Team	Home Win %	Draw %	Predicted
Jun 11	Mexico	South Africa	64.7%	23.2%	Mexico Win
Jun 11	South Korea	Czech Republic	47.1%	24.5%	South Korea Win
Jun 12	Canada	Bosnia and Herzegovina	70.4%	17.8%	Canada Win
Jun 12	United States	Paraguay	34.0%	28.1%	Paraguay Win
Jun 13	Qatar	Switzerland	10.6%	13.6%	Switzerland Win
Jun 13	Brazil	Morocco	46.1%	29.1%	Brazil Win
Jun 13	Haiti	Scotland	28.7%	24.3%	Scotland Win
Jun 13	Australia	Turkey	34.7%	29.1%	Turkey Win

5.2 Second Matchday (June 14-17, 2026)

Date	Home Team	Away Team	Home Win %	Draw %	Predicted
Jun 14	Germany	Curacao	74.0%	21.5%	Germany Win
Jun 14	Ivory Coast	Ecuador	21.2%	26.5%	Ecuador Win
Jun 14	Netherlands	Japan	38.5%	27.2%	Netherlands Win
Jun 14	Sweden	Tunisia	46.7%	25.9%	Sweden Win

Jun 15	Belgium	Egypt	54.0%	28.6%	Belgium Win
Jun 15	Iran	New Zealand	49.3%	28.5%	Iran Win
Jun 15	Spain	Cape Verde	91.0%	6.6%	Spain Win
Jun 15	Saudi Arabia	Uruguay	12.9%	19.3%	Uruguay Win
Jun 16	France	Senegal	64.4%	17.7%	France Win
Jun 16	Iraq	Norway	24.4%	23.1%	Norway Win
Jun 16	Argentina	Algeria	70.3%	23.1%	Argentina Win
Jun 16	Austria	Jordan	45.7%	25.2%	Austria Win
Jun 17	Portugal	DR Congo	57.0%	26.9%	Portugal Win
Jun 17	Uzbekistan	Colombia	19.3%	26.7%	Colombia Win
Jun 17	England	Croatia	57.7%	20.4%	England Win
Jun 17	Ghana	Panama	22.4%	25.5%	Panama Win

5.3 Third Matchday (June 18-20, 2026)

Date	Home Team	Away Team	Home Win %	Draw %	Predicted
Jun 18	Mexico	South Korea	47.6%	28.9%	Mexico Win
Jun 18	Czech Republic	South Africa	56.4%	24.5%	Czech Republic Win
Jun 18	Switzerland	Bosnia and Herzegovina	73.0%	16.6%	Switzerland Win
Jun 18	Canada	Qatar	75.7%	16.6%	Canada Win
Jun 19	Scotland	Morocco	20.4%	25.5%	Morocco Win
Jun 19	Brazil	Haiti	74.6%	20.0%	Brazil Win
Jun 19	United States	Australia	31.8%	29.2%	Australia Win
Jun 19	Turkey	Paraguay	40.0%	27.1%	Turkey Win
Jun 20	Germany	Ivory Coast	53.7%	31.0%	Germany Win
Jun 20	Ecuador	Curacao	81.1%	14.1%	Ecuador Win

Jun 20	Netherlands	Sweden	59.8%	21.7%	Netherlands Win
Jun 20	Tunisia	Japan	11.6%	23.4%	Japan Win

5.4 Final Group Stage Matchday (June 21-27, 2026)

Date	Home Team	Away Team	Home Win %	Draw %	Predicted
Jun 21	Belgium	Iran	40.9%	25.6%	Belgium Win
Jun 21	New Zealand	Egypt	33.6%	33.2%	Egypt Win
Jun 21	Spain	Saudi Arabia	80.7%	14.8%	Spain Win
Jun 21	Uruguay	Cape Verde	80.8%	15.7%	Uruguay Win
Jun 22	France	Iraq	77.9%	16.4%	France Win
Jun 22	Norway	Senegal	41.3%	20.8%	Norway Win
Jun 22	Argentina	Austria	69.8%	23.2%	Argentina Win
Jun 22	Jordan	Algeria	31.4%	25.8%	Algeria Win
Jun 23	Portugal	Uzbekistan	59.9%	27.3%	Portugal Win
Jun 23	Colombia	DR Congo	67.1%	22.7%	Colombia Win
Jun 23	England	Ghana	85.3%	11.2%	England Win
Jun 23	Panama	Croatia	25.3%	28.1%	Croatia Win
Jun 24	Mexico	Czech Republic	51.5%	30.0%	Mexico Win
Jun 24	South Africa	South Korea	21.5%	26.9%	South Korea Win
Jun 24	Canada	Switzerland	34.2%	29.9%	Switzerland Win
Jun 24	Bosnia and Herzegovina	Qatar	55.4%	19.3%	Bosnia Win
Jun 24	Scotland	Brazil	14.0%	24.1%	Brazil Win
Jun 24	Morocco	Haiti	69.7%	20.5%	Morocco Win
Jun 25	United States	Turkey	31.6%	27.9%	Turkey Win

Jun 25	Paraguay	Australia	34.3%	30.1%	Australia Win
Jun 25	Curacao	Ivory Coast	26.2%	25.9%	Ivory Coast Win
Jun 25	Ecuador	Germany	41.0%	27.9%	Ecuador Win
Jun 25	Japan	Sweden	56.0%	24.9%	Japan Win
Jun 25	Tunisia	Netherlands	13.0%	23.8%	Netherlands Win
Jun 26	Egypt	Iran	28.9%	26.2%	Iran Win
Jun 26	New Zealand	Belgium	26.0%	27.9%	Belgium Win
Jun 26	Cape Verde	Saudi Arabia	33.6%	29.1%	Saudi Arabia Win
Jun 26	Uruguay	Spain	22.2%	32.0%	Spain Win
Jun 26	Norway	France	25.9%	21.9%	France Win
Jun 26	Senegal	Iraq	60.2%	23.3%	Senegal Win
Jun 27	Algeria	Austria	31.5%	27.6%	Austria Win
Jun 27	Jordan	Argentina	9.0%	15.4%	Argentina Win
Jun 27	Colombia	Portugal	40.8%	31.0%	Colombia Win
Jun 27	DR Congo	Uzbekistan	30.9%	28.4%	Uzbekistan Win
Jun 27	Panama	England	10.0%	20.0%	England Win
Jun 27	Croatia	Ghana	77.4%	17.1%	Croatia Win

5.5 Notable Predictions and Observations

Looking across all 72 group stage matches, a few patterns and interesting individual predictions stand out.

- **Spain vs Cape Verde** on June 15 produces the most one-sided prediction of the entire tournament, with Spain given a 91% chance of winning. Given Spain's ELO rating of 1617 compared to Cape Verde's 1130, this is not surprising. Only about 2.4% chance is given to Cape Verde winning outright.
- **Jordan vs Argentina** on June 27 is predicted even more lopsidedly in Argentina's favour, with Argentina having a 75.6% win probability and Jordan just 9%. Argentina's ELO of 1630 is nearly 400 points higher than Jordan's 1259.

- **United States vs Paraguay** on June 12 is predicted to go Paraguay's way at 38%, with the US given just a 34% chance of winning on home soil. This is one of the bigger potential surprises of the early rounds.
- **Ecuador vs Germany** on June 25 is predicted as an Ecuador win at 41%, which would be a significant upset given Germany's higher ELO. The probabilities are close enough, however, that this is genuinely a toss-up: Germany has a 31% chance of winning.
- **Brazil's group** looks manageable on paper, with wins predicted against Morocco, Haiti, and Scotland. However, the Morocco match on June 13 is particularly close, with Brazil at only 46.1% and Morocco at 24.7%, which makes it one of the more interesting early contests.

6. Key Insights from 154 Years of Football

6.1 Home Advantage Is Real and Significant

Of the 49,215 matches in the dataset, 49% were won by the home team, compared to only 28.3% won by the away team and 22.7% drawn. This is not a small margin. Home advantage is a concrete, measurable phenomenon in international football, and the analysis confirms that teams genuinely perform better when playing in familiar conditions in front of their own supporters.

6.2 The Top Teams Have Been Consistent for Decades

Argentina, Brazil, Germany, England, and Spain have all been near the top of the global rankings for generations. Their high ELO ratings are not accidents of recent form but the product of sustained excellence over many decades and thousands of matches. This consistency is itself a signal: the structural factors that make a country good at football, including investment in youth development, coaching quality, and the strength of domestic leagues, tend to be durable.

6.3 Win Rate Alone Does Not Tell the Full Story

Brazil's win rate of 63.4% is the highest among major nations, yet they rank sixth in ELO behind Argentina, Spain, France, England, and Colombia. This is because ELO weights the quality of the opponent. Winning 10 friendlies against weaker nations looks different in ELO terms from losing a World Cup semi-final to a top-five team. The ELO system is more sophisticated than a simple win percentage and provides a more accurate picture of a team's true standing.

6.4 Football Has Changed Over Time

The jump from 127 matches in the Pioneer Era to over 16,000 matches in the Globalization Era tells its own story. International football was once a niche activity played mostly by a handful of European nations. It is now a truly global sport with 333 teams represented in the dataset, from Argentina to American Samoa, from Germany to Guam. The sport has expanded both geographically and in terms of the sheer volume of competitive fixtures.

6.5 Machine Learning Improves on Simple Rules

The improvement from a 47.2% baseline accuracy to 58.2% with XGBoost may not sound dramatic in absolute terms, but it represents a 23% relative improvement. It also shows that there are genuine, learnable patterns in football data beyond simply 'home teams win more often'. Recent form, goal scoring history, and the quality gap between opponents all contribute meaningfully to prediction accuracy.

7. Methodology Summary

For readers who want to understand the technical side of the project without diving into code, this section provides a plain-language walkthrough of how each analytical step was carried out.

7.1 Data Collection and Cleaning

The raw data was taken from publicly available international football records. The cleaning process involved three main tasks. First, team names were standardized using the former names file to ensure historical continuity. Second, tournaments were classified into tiers to reflect their competitive importance. Third, match records were checked for completeness and any obvious errors were addressed. The output of this stage is the clean_matches.csv and clean_goalscorers.csv files.

7.2 ELO Calculation

The ELO calculation was applied sequentially, processing matches in chronological order from 1872 to the present. Each match updates the ratings of both teams based on the result. The K-factor, which controls how much each match affects ratings, was set higher for World Cup and major tournament matches and lower for friendlies. Home advantage was incorporated as a fixed bonus to the expected win probability for the home team. After all matches were processed, each team's final rating was saved, along with a history of how their rating changed over time.

7.3 Machine Learning Model

The machine learning model uses XGBoost, which is a powerful and widely used algorithm that builds predictions by combining many small decision trees. Each tree looks at a subset of the features and makes a simple rule-based prediction. The final prediction is a weighted combination of all these small predictions, which allows the model to capture complex interactions between variables.

The model was trained on historical matches and tested on a held-out set of matches it had not seen during training. The target variable is the match result: home win, draw, or away win. The model outputs a probability for each of the three outcomes, and the highest probability determines the predicted result.

7.4 Predicting 2026 World Cup Matches

For the 2026 World Cup group stage predictions, the model was run on each upcoming fixture using the current ELO ratings and contextual information for each match. Because these matches had not yet taken place at the time of analysis, pre-match ELO ratings were used, and recent form statistics were taken from each team's most recent historical matches in the dataset.

8. Conclusion

This project demonstrates what becomes possible when you combine a rich historical dataset with thoughtful analytical methods. Starting from over 49,000 matches spanning 154 years, the analysis has produced a comprehensive picture of international football: which teams are truly the best in the world, what factors most influence match outcomes, and what the data suggests will happen in the 2026 FIFA World Cup group stage.

The ELO rating system proves to be a robust and fair way to rank international teams. Argentina's position at the top of the rankings, with a rating of 1629.63, reflects their consistent excellence across decades of competition and their status as reigning world champions. Spain, France, England, and Brazil round out the tier of elite teams whose class is beyond dispute.

The machine learning model, achieving 58.2% accuracy with XGBoost, represents a meaningful advance over simply assuming the home team always wins. The most important predictors turn out to be ELO-based measures of team quality, confirming that long-term track record is the best guide to future performance. But recent form, goal-scoring patterns, and match context all contribute meaningfully.

For the 2026 World Cup group stage, the model points to a set of predictable outcomes in matches involving the very top teams. Spain, Argentina, France, England, and Brazil are all predicted to win their group stage matches comfortably. But the model also identifies genuinely competitive match-ups, such as the United States versus Paraguay and Ecuador versus Germany, where the data suggests the underdog has a real chance.

Football, of course, will always find a way to surprise us. That is part of what makes it the world's game. But this project shows that even in a sport defined by unpredictability, the data has a great deal to tell us if we take the time to listen.